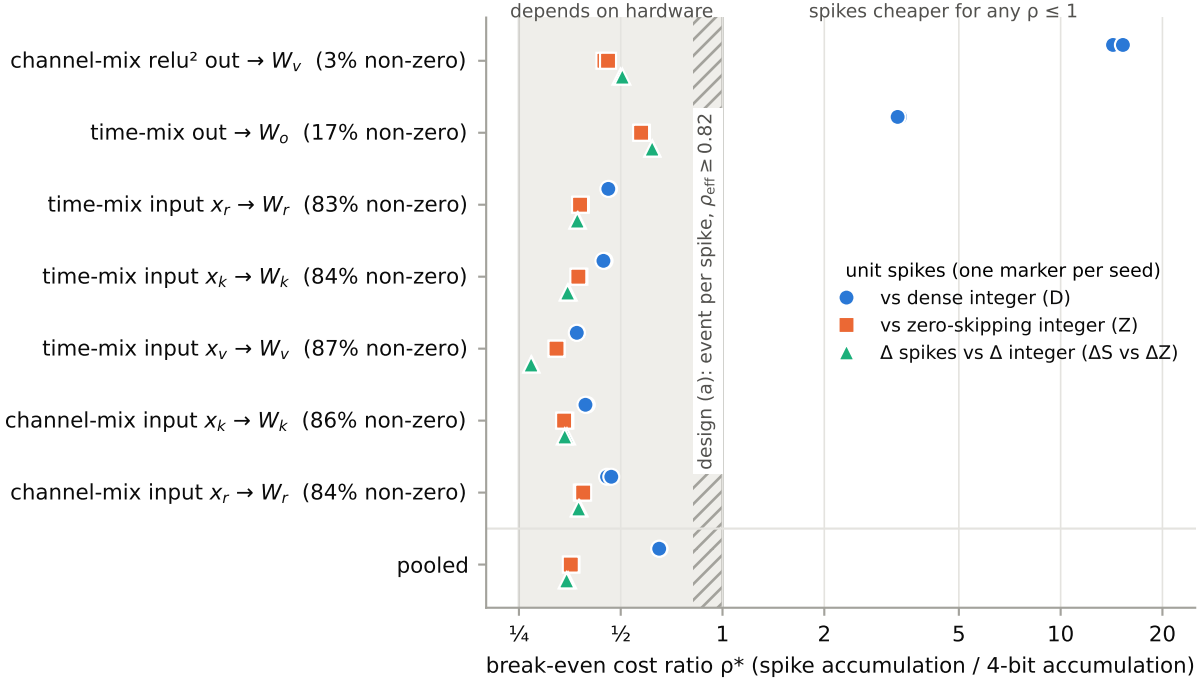
depends on hardware
spikes cheaper for any ρ ≤ 1
channel-mix relu² out → W_v (3% non-zero)
time-mix out → W_o (17% non-zero)
time-mix input x_r → W_r (83% non-zero)
time-mix input x_k → W_k (84% non-zero)
time-mix input x_v → W_v (87% non-zero)
channel-mix input x_k → W_k (86% non-zero)
channel-mix input x_r → W_r (84% non-zero)
pooled
design (a): event per spike, $\rho_{eff} \geq 0.82$
unit spikes (one marker per seed)
vs dense integer (D)
vs zero-skipping integer (Z)
Δ spikes vs Δ integer (ΔS vs ΔZ)
¼
½
1
2
5
10
20
break-even cost ratio ρ* (spike accumulation / 4-bit accumulation)